

Geospatial Assessment of Wetland Dynamics and Runoff for Sustainable Watershed Management: A Case Study of the Meghadri Gedda Reservoir, Visakhapatnam, Andhra Pradesh, India

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Due to the rapid urbanization, depletion of groundwater, and climate variability, sustainable watershed management is a growing pressure. This paper uses a combination of Remote Sensing (RS) and Geographic Information Systems (GIS) to examine hydrological dynamics, morphometrics, and land use dynamics within the Meghadri gedda watershed (366 km²) of the Visakhapatnam district, India. Satellite data Landsat (1976, 2005, 2010, 2011, 2015) and Sentinel-2 (2020) was used to obtain multi-temporal data sets to be used in land use/land cover (LULC) and change detection classification. The ArcGIS was used to create thematic layers such as drainage, slope, soil groups, hydrogeomorphology, NDVI, and NDWI. The SCS-Curve Number (SCS-CN) model was used to estimate surface runoff through a combination of soil parameters, land use parameters and terrain parameters. Findings suggest that the built-up area has grown significantly whereas surface water bodies and vegetation cover have significantly diminished over the last 40 years. According to the estimation of runoff, there is greater surface runoff over the past years, which is due to the increase of the impervious surfaces and tank siltation. Change detection also validates depleting groundwater recharge capability owing to deterioration of natural storage characteristic. The geospatial analysis of suitability proposed thirteen of the best locations to install the check dams and twelve recharge pits to desilt and augment the groundwater. The socio- economic value of the watershed is highlighted by the Meghadri gedda reservoir, which serves almost 8 million gallons of water daily to Visakhapatnam city. The paper has shown that, integrated RS-GIS modelling offers cost-efficient and scalable decision support model in sustainable watershed planning and climate-resilient water management.

Keywords: Morphometric analysis, Remote sensing, SCS-CN method, Hydrological modelling and Watershed Management.